Classification Tree

Libraries

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✓ dplyr     1.2.1     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.3     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
 conflicts to become errors
```

```
library(tidymodels)
```

```
## ── Attaching packages ─────────────────────────────────── tidymodels 1.
5.0 ──
## ✓ broom        1.0.12     ✓ rsample      1.3.2
## ✓ dials        1.4.3      ✓ tailor       0.1.0
## ✓ infer        1.1.0      ✓ tune         2.1.0
## ✓ modeldata    1.5.1      ✓ workflows    1.3.0
## ✓ parsnip      1.6.0      ✓ workflowsets 1.1.1
## ✓ recipes      1.3.2      ✓ yardstick    1.4.0
## ── Conflicts ────────────────────────────────────── tidymodels_conflict
s() ──
## ✖ scales::discard() masks purrr::discard()
## ✖ dplyr::filter()   masks stats::filter()
## ✖ recipes::fixed()  masks stringr::fixed()
## ✖ dplyr::lag()      masks stats::lag()
## ✖ yardstick::spec() masks readr::spec()
## ✖ recipes::step()   masks stats::step()
```

```
library(rpart)
```

```
##
## Attaching package: 'rpart'
##
## The following object is masked from 'package:dials':
##
##     prune
```

```
library(rpart.plot)
library(RColorBrewer)
library(rattle)
```

```
## Loading required package: bitops
## Rattle: A free graphical interface for data science with R.
## Version 5.6.2 Copyright (c) 2006-2023 Togaware Pty Ltd.
## Type 'rattle()' to shake, rattle, and roll your data.
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following objects are masked from 'package:yardstick':
##
##     precision, recall, sensitivity, specificity
##
## The following object is masked from 'package:rsample':
##
##     calibration
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(skimr)
library(mice)
```

```
##
## Attaching package: 'mice'
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following objects are masked from 'package:base':
##
##     cbind, rbind
```

Load dataset

```
train <- read_csv("C:/Users/miche/OneDrive/Desktop/BAN 502 Predictive Analyti
cs/Course Project/train.csv")
```

```
## Rows: 26570 Columns: 26
## ── Column specification ────────────────────────────────────────────────────
──────
## Delimiter: ","
## chr  (4): product_code, attribute_0, attribute_1, failure
## dbl (22): id, loading, attribute_2, attribute_3, measurement_0, measuremen
```

```
t_...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.
```

Structure and summary

```
str(train)
```

```
## spc_tbl_ [26,570 × 26] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ id             : num [1:26570] 0 1 2 3 4 5 6 7 8 9 ...
##  $ product_code   : chr [1:26570] "A" "A" "A" "A" ...
##  $ loading        : num [1:26570] 80.1 84.9 82.4 101.1 188.1 ...
##  $ attribute_0    : chr [1:26570] "material_7" "material_7" "material_7" "m
aterial_7" ...
##  $ attribute_1    : chr [1:26570] "material_8" "material_8" "material_8" "m
aterial_8" ...
##  $ attribute_2    : num [1:26570] 9 9 9 9 9 9 9 9 9 9 ...
##  $ attribute_3    : num [1:26570] 5 5 5 5 5 5 5 5 5 5 ...
##  $ measurement_0  : num [1:26570] 7 14 12 13 9 11 12 4 9 10 ...
##  $ measurement_1  : num [1:26570] 8 3 1 2 2 4 2 8 6 4 ...
##  $ measurement_2  : num [1:26570] 4 3 5 6 8 0 4 8 5 7 ...
##  $ measurement_3  : num [1:26570] 18 18.2 18.1 17.3 19.3 ...
##  $ measurement_4  : num [1:26570] 12.5 11.5 11.7 11.2 12.9 ...
##  $ measurement_5  : num [1:26570] 15.7 17.7 16.7 18.6 17 ...
##  $ measurement_6  : num [1:26570] 19.3 17.9 18.2 18.3 15.7 ...
##  $ measurement_7  : num [1:26570] 11.7 12.7 12.7 12.6 11.3 ...
##  $ measurement_8  : num [1:26570] 20.2 17.9 18.3 19.1 18.1 ...
##  $ measurement_9  : num [1:26570] 10.7 12.4 12.7 12.5 10.3 ...
##  $ measurement_10: num [1:26570] 15.9 17.9 15.6 16.3 17.1 ...
##  $ measurement_11: num [1:26570] 17.6 17.9 NA 18.4 19.9 ...
##  $ measurement_12: num [1:26570] 15.2 11.8 13.8 10 12.4 ...
##  $ measurement_13: num [1:26570] 15 14.7 16.7 15.2 16.2 ...
##  $ measurement_14: num [1:26570] NA 15.4 18.6 15.6 12.8 ...
##  $ measurement_15: num [1:26570] 13 14.4 14.1 16.2 13.2 ...
##  $ measurement_16: num [1:26570] 14.7 15.6 17.9 17.2 16.4 ...
##  $ measurement_17: num [1:26570] 764 682 663 826 580 ...
##  $ failure        : chr [1:26570] "No" "No" "No" "No" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   id = col_double(),
##   ..   product_code = col_character(),
##   ..   loading = col_double(),
##   ..   attribute_0 = col_character(),
##   ..   attribute_1 = col_character(),
##   ..   attribute_2 = col_double(),
##   ..   attribute_3 = col_double(),
##   ..   measurement_0 = col_double(),
##   ..   measurement_1 = col_double(),
```

```
##   ..   measurement_2 = col_double(),
##   ..   measurement_3 = col_double(),
##   ..   measurement_4 = col_double(),
##   ..   measurement_5 = col_double(),
##   ..   measurement_6 = col_double(),
##   ..   measurement_7 = col_double(),
##   ..   measurement_8 = col_double(),
##   ..   measurement_9 = col_double(),
##   ..   measurement_10 = col_double(),
##   ..   measurement_11 = col_double(),
##   ..   measurement_12 = col_double(),
##   ..   measurement_13 = col_double(),
##   ..   measurement_14 = col_double(),
##   ..   measurement_15 = col_double(),
##   ..   measurement_16 = col_double(),
##   ..   measurement_17 = col_double(),
##   ..   failure = col_character()
##   .. )
##  - attr(*, "problems")=<pointer: 0x000001f8212ff110>

summary(train)

##        id           product_code      loading          attribute_0
##  Min.   :    0   Length   :26570   Min.   : 33.16   Length   :26570
##  1st Qu.: 6642   N.unique :    5   1st Qu.: 99.99   N.unique :    2
##  Median :13284   N.blank  :    0   Median :122.39   N.blank  :    0
##  Mean   :13284   Min.nchar:    1   Mean   :127.83   Min.nchar:   10
##  3rd Qu.:19927   Max.nchar:    1   3rd Qu.:149.15   Max.nchar:   10
##  Max.   :26569                     Max.   :385.86
##                                    NAs    :250
##     attribute_1     attribute_2     attribute_3   measurement_0
##  Length   :26570   Min.   :5.000   Min.   :5.00   Min.   : 0.000
##  N.unique :    3   1st Qu.:6.000   1st Qu.:6.00   1st Qu.: 4.000
##  N.blank  :    0   Median :6.000   Median :8.00   Median : 7.000
##  Min.nchar:   10   Mean   :6.754   Mean   :7.24   Mean   : 7.416
##  Max.nchar:   10   3rd Qu.:8.000   3rd Qu.:8.00   3rd Qu.:10.000
##                    Max.   :9.000   Max.   :9.00   Max.   :29.000
##
##  measurement_1    measurement_2    measurement_3   measurement_4
##  Min.   : 0.000   Min.   : 0.000   Min.   :13.97   Min.   : 8.008
##  1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:17.12   1st Qu.:11.051
##  Median : 8.000   Median : 6.000   Median :17.79   Median :11.733
##  Mean   : 8.233   Mean   : 6.257   Mean   :17.79   Mean   :11.732
##  3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:18.47   3rd Qu.:12.410
##  Max.   :29.000   Max.   :24.000   Max.   :21.50   Max.   :16.484
##                                    NAs    :381     NAs    :538
##  measurement_5   measurement_6   measurement_7    measurement_8
##  Min.   :12.07   Min.   :12.71   Min.   : 7.968   Min.   :15.22
##  1st Qu.:16.44   1st Qu.:16.84   1st Qu.:11.045   1st Qu.:18.34
##  Median :17.13   Median :17.52   Median :11.712   Median :19.02
```

```
##  Mean   :17.13   Mean   :17.51   Mean   :11.717   Mean   :19.02
##  3rd Qu.:17.80   3rd Qu.:18.18   3rd Qu.:12.391   3rd Qu.:19.71
##  Max.   :21.43   Max.   :21.54   Max.   :15.419   Max.   :23.81
##  NAs    :676     NAs    :796     NAs    :937     NAs    :1048
##  measurement_9    measurement_10   measurement_11  measurement_12
##  Min.   : 7.537   Min.   : 9.323   Min.   :12.46   Min.   : 5.167
##  1st Qu.:10.757   1st Qu.:15.209   1st Qu.:18.17   1st Qu.:10.703
##  Median :11.430   Median :16.127   Median :19.21   Median :11.717
##  Mean   :11.431   Mean   :16.118   Mean   :19.17   Mean   :11.702
##  3rd Qu.:12.102   3rd Qu.:17.025   3rd Qu.:20.21   3rd Qu.:12.709
##  Max.   :15.412   Max.   :22.479   Max.   :25.64   Max.   :17.663
##  NAs    :1227     NAs    :1300     NAs    :1468    NAs    :1601
##  measurement_13  measurement_14  measurement_15   measurement_16
##  Min.   :10.89   Min.   : 9.14   Min.   : 9.104   Min.   : 9.701
##  1st Qu.:14.89   1st Qu.:15.06   1st Qu.:13.957   1st Qu.:15.268
##  Median :15.63   Median :16.04   Median :14.969   Median :16.436
##  Mean   :15.65   Mean   :16.05   Mean   :14.996   Mean   :16.461
##  3rd Qu.:16.37   3rd Qu.:17.08   3rd Qu.:16.018   3rd Qu.:17.628
##  Max.   :22.71   Max.   :22.30   Max.   :21.626   Max.   :24.094
##  NAs    :1774    NAs    :1874    NAs    :2009    NAs    :2110
##  measurement_17        failure
##  Min.   : 196.8   Length   :26570
##  1st Qu.: 619.0   N.unique :    2
##  Median : 701.0   N.blank  :    0
##  Mean   : 701.3   Min.nchar:    2
##  3rd Qu.: 784.1   Max.nchar:    3
##  Max.   :1312.8
##  NAs    :2284
```

Delete ID column

```
train = train %>% select(- id)
```

Convert character variables to factors

```
train = train %>% mutate_if(is.character, as_factor)
```

Convert other categorical variables to factors

```
train <- train %>%
  mutate(
    attribute_2 = as.factor(attribute_2),
    attribute_3 = as.factor(attribute_3)
  )
```

Clean data, remove outliers

```
train = train %>% filter(measurement_2 < 22)
train = train %>% filter(measurement_8 < 23.5)
train = train %>% filter(measurement_14 > 8)
```

```
train = train %>% filter(measurement_15 < 23)
train = train %>% filter(measurement_17 < 1500)
```

Check for missing data

```
skim(train)
```

Data summary

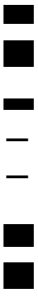
Name	train
Number of rows	20043
Number of columns	25
_____	
Column type frequency:	
factor	6
numeric	19
_____	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
product_code	0	1	FALSE	5	C: 4359, B: 3992, E: 3978, A: 3905
attribute_0	0	1	FALSE	2	mat: 16051, mat: 3992
attribute_1	0	1	FALSE	3	mat: 8264, mat: 7801, mat: 3978
attribute_2	0	1	FALSE	4	6: 7787, 5: 4359, 8: 3992, 9: 3905
attribute_3	0	1	FALSE	4	8: 8351, 9: 3978, 5: 3905, 6: 3809
failure	0	1	FALSE	2	No: 15803, Yes: 4240

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
loading	178	0.99	127.88	38.94	40.81	100.16	122.48	149.15	385.86	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_0	0	1.00	7.43	4.15	0.00	4.00	7.00	10.00	27.00	
measurement_1	0	1.00	8.21	4.20	0.00	5.00	8.00	11.00	29.00	
measurement_2	0	1.00	6.25	3.30	0.00	4.00	6.00	8.00	21.00	
measurement_3	280	0.99	17.79	1.00	14.10	17.11	17.79	18.47	21.34	
measurement_4	423	0.98	11.73	0.99	8.01	11.05	11.73	12.40	16.48	
measurement_5	501	0.98	17.13	1.00	12.07	16.45	17.13	17.81	21.42	
measurement_6	599	0.97	17.51	1.00	12.71	16.83	17.52	18.17	21.54	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_7	684	0.97	11.72	1.00	7.97	11.05	11.72	12.39	15.35	
measurement_8	0	1.00	19.02	1.01	15.27	18.34	19.02	19.71	23.33	
measurement_9	930	0.95	11.43	1.00	7.54	10.76	11.43	12.10	15.15	
measurement_10	980	0.95	16.13	1.41	9.32	15.21	16.14	17.03	22.48	
measurement_11	1066	0.95	19.18	1.52	12.46	18.16	19.23	20.21	25.64	
measurement_12	1191	0.94	11.71	1.49	5.17	10.70	11.73	12.72	17.59	
measurement_13	1369	0.93	15.65	1.16	10.89	14.88	15.63	16.38	22.71	
measurement_14	0	1.00	16.05	1.48	9.95	15.06	16.05	17.08	22.30	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_15	0	1.00	15.00	1.55	9.10	13.95	14.98	16.03	21.52	
measurement_16	1575	0.92	16.47	1.71	9.70	15.27	16.45	17.63	24.09	
measurement_17	0	1.00	701.26	123.05	196.79	618.64	700.76	783.41	1312.79	

Imputation

```
set.seed(1234)
imp_train = mice(train, m=5, method="pmm", printFlag=FALSE)

## Warning: Number of logged events: 594

train_complete = complete(imp_train)

skim(train_complete)
```

Data summary

Name	train_complete
Number of rows	20043
Number of columns	25
_____	
Column type frequency:	
factor	6
numeric	19
_____	
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
product_code	0	1	FALSE	5	C: 4359, B: 3992, E: 3978, A: 3905
attribute_0	0	1	FALSE	2	mat: 16051, mat: 3992
attribute_1	0	1	FALSE	3	mat: 8264, mat: 7801, mat: 3978
attribute_2	0	1	FALSE	4	6: 7787, 5: 4359, 8: 3992, 9: 3905
attribute_3	0	1	FALSE	4	8: 8351, 9: 3978, 5: 3905, 6: 3809
failure	0	1	FALSE	2	No: 15803, Yes: 4240

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
loading	0	1	127.82	39.41	40.81	99.76	122.25	149.12	385.86	
measurement_0	0	1	7.43	4.15	0.00	4.00	7.00	10.00	27.00	
measurement_1	0	1	8.21	4.20	0.00	5.00	8.00	11.00	29.00	
measurement_2	0	1	6.25	3.30	0.00	4.00	6.00	8.00	21.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_3	0	1	17.79	1.00	14.10	17.11	17.78	18.48	21.34	
measurement_4	0	1	11.74	0.99	8.01	11.07	11.76	12.40	16.48	
measurement_5	0	1	17.17	1.02	12.07	16.47	17.16	17.87	21.42	
measurement_6	0	1	17.46	1.07	12.71	16.80	17.51	18.15	21.54	
measurement_7	0	1	11.65	1.07	7.97	10.98	11.67	12.37	15.35	
measurement_8	0	1	19.02	1.01	15.27	18.34	19.02	19.71	23.33	
measurement_9	0	1	11.43	1.01	7.54	10.74	11.45	12.12	15.15	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_10	0	1	16.23	1.51	9.32	15.22	16.19	17.15	22.48	
measurement_11	0	1	19.07	1.65	12.46	18.06	19.16	20.20	25.64	
measurement_12	0	1	11.72	1.48	5.17	10.77	11.70	12.71	17.59	
measurement_13	0	1	15.59	1.15	10.89	14.80	15.57	16.31	22.71	
measurement_14	0	1	16.05	1.48	9.95	15.06	16.05	17.08	22.30	
measurement_15	0	1	15.00	1.55	9.10	13.95	14.98	16.03	21.52	
measurement_16	0	1	16.75	2.10	9.70	15.29	16.57	17.89	24.09	
measurement_17	0	1	701.26	123.05	196.79	618.64	700.76	783.41	1312.79	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
										▇▁▁

Structure train_complete

```
str(train_complete)
```

```
## 'data.frame':    20043 obs. of  25 variables:
##  $ product_code  : Factor w/ 5 levels "A","B","C","D",..: 1 1 1 1 1 1 1 1
1 1 ...
##  $ loading       : num  84.9 82.4 101.1 188.1 177.9 ...
##  $ attribute_0   : Factor w/ 2 levels "material_7","material_5": 1 1 1 1 1
 1 1 1 1 1 ...
##  $ attribute_1   : Factor w/ 3 levels "material_8","material_5",..: 1 1 1
1 1 1 1 1 1 1 ...
##  $ attribute_2   : Factor w/ 4 levels "5","6","8","9": 4 4 4 4 4 4 4 4 4 4
 ...
##  $ attribute_3   : Factor w/ 4 levels "5","6","8","9": 1 1 1 1 1 1 1 1 1 1
 ...
##  $ measurement_0 : num  14 12 13 9 4 10 7 9 6 10 ...
##  $ measurement_1 : num  3 1 2 2 8 3 3 0 9 4 ...
##  $ measurement_2 : num  3 5 6 8 8 6 2 6 10 10 ...
##  $ measurement_3 : num  18.2 18.1 17.3 19.3 17.1 ...
##  $ measurement_4 : num  11.5 11.7 11.2 12.9 13.6 ...
##  $ measurement_5 : num  17.7 16.7 18.6 17 17.9 ...
##  $ measurement_6 : num  17.9 18.2 18.3 15.7 15.9 ...
##  $ measurement_7 : num  12.7 12.7 12.6 11.3 11 ...
##  $ measurement_8 : num  17.9 18.3 19.1 18.1 18.6 ...
##  $ measurement_9 : num  12.4 12.7 12.5 10.3 10.3 ...
##  $ measurement_10: num  17.9 15.6 16.3 17.1 16.4 ...
##  $ measurement_11: num  17.9 19 18.4 19.9 20.5 ...
##  $ measurement_12: num  11.8 13.8 10 12.4 12.2 ...
##  $ measurement_13: num  14.7 16.7 15.2 16.2 15.6 ...
##  $ measurement_14: num  15.4 18.6 15.6 12.8 17 ...
##  $ measurement_15: num  14.4 14.1 16.2 13.2 15.2 ...
##  $ measurement_16: num  15.6 17.9 17.2 16.4 17.2 ...
##  $ measurement_17: num  682 663 826 580 684 ...
##  $ failure       : Factor w/ 2 levels "No","Yes": 1 1 1 1 2 1 2 1 2 1 ...
```

Split data

```
set.seed(123)
train_complete_split = initial_split(train_complete, prop = 0.7, strata = fai
lure)
train_set = training(train_complete_split)
test_set = testing(train_complete_split)
```

structure of train_set

```
summary(train_set)
```

```
##  product_code    loading           attribute_0        attribute_1   attrib
ute_2
##  A:2746       Min.   : 40.81   material_7:11232   material_8:5786   5:3040
##  B:2798       1st Qu.: 99.73   material_5: 2798   material_5:5483   6:5446
##  C:3040       Median :122.20                      material_6:2761   8:2798
##  D:2685       Mean   :127.78                                        9:2746
##  E:2761       3rd Qu.:149.19
##               Max.   :385.86
##  attribute_3 measurement_0    measurement_1    measurement_2    measuremen
t_3
##  5:2746      Min.   : 0.000   Min.   : 0.000   Min.   : 0.000   Min.   :1
4.10
##  6:2685      1st Qu.: 4.000   1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:1
7.12
##  8:5838      Median : 7.000   Median : 8.000   Median : 6.000   Median :1
7.79
##  9:2761      Mean   : 7.435   Mean   : 8.216   Mean   : 6.275   Mean   :1
7.79
##              3rd Qu.:10.000   3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:1
8.48
##              Max.   :27.000   Max.   :29.000   Max.   :21.000   Max.   :2
1.22
##  measurement_4    measurement_5   measurement_6   measurement_7
##  Min.   : 8.097   Min.   :12.81   Min.   :12.71   Min.   : 7.973
##  1st Qu.:11.078   1st Qu.:16.46   1st Qu.:16.81   1st Qu.:10.981
##  Median :11.762   Median :17.17   Median :17.51   Median :11.673
##  Mean   :11.742   Mean   :17.17   Mean   :17.46   Mean   :11.651
##  3rd Qu.:12.403   3rd Qu.:17.87   3rd Qu.:18.15   3rd Qu.:12.371
##  Max.   :16.484   Max.   :21.43   Max.   :21.33   Max.   :15.352
##  measurement_8   measurement_9    measurement_10   measurement_11
##  Min.   :15.27   Min.   : 7.537   Min.   : 9.323   Min.   :12.46
##  1st Qu.:18.34   1st Qu.:10.737   1st Qu.:15.219   1st Qu.:18.07
##  Median :19.02   Median :11.439   Median :16.181   Median :19.16
##  Mean   :19.02   Mean   :11.427   Mean   :16.223   Mean   :19.08
##  3rd Qu.:19.71   3rd Qu.:12.122   3rd Qu.:17.148   3rd Qu.:20.20
##  Max.   :22.75   Max.   :15.099   Max.   :21.696   Max.   :25.64
##  measurement_12   measurement_13  measurement_14   measurement_15
##  Min.   : 5.167   Min.   :11.04   Min.   : 9.947   Min.   : 9.255
##  1st Qu.:10.765   1st Qu.:14.80   1st Qu.:15.065   1st Qu.:13.965
```

```
##  Median :11.691   Median :15.57   Median :16.050   Median :14.985
##  Mean   :11.719   Mean   :15.59   Mean   :16.054   Mean   :15.009
##  3rd Qu.:12.705   3rd Qu.:16.31   3rd Qu.:17.086   3rd Qu.:16.040
##  Max.   :17.594   Max.   :22.71   Max.   :22.303   Max.   :21.515
##  measurement_16   measurement_17   failure
##  Min.   : 9.701   Min.   : 206.6   No :11062
##  1st Qu.:15.275   1st Qu.: 618.8   Yes: 2968
##  Median :16.567   Median : 701.4
##  Mean   :16.744   Mean   : 701.8
##  3rd Qu.:17.879   3rd Qu.: 784.7
##  Max.   :24.094   Max.   :1312.8
```

Build classification tree

```
train_recipe = recipe(failure ~ loading, train_set) %>%
  step_dummy(all_nominal(), -all_outcomes())

tree_model = decision_tree() %>%
  set_engine("rpart", model = TRUE) %>%
  set_mode("classification")

train_wflow =
  workflow() %>%
  add_model(tree_model) %>%
  add_recipe(train_recipe)

train_fit = fit(train_wflow, train_set)
```

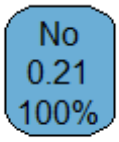
Assign tree

```
tree = train_fit %>%
  pull_workflow_fit() %>%
  pluck("fit")

## Warning: `pull_workflow_fit()` was deprecated in workflows 0.2.3.
## ℹ Please use `extract_fit_parsnip()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Visualize tree

```
rpart.plot(tree)
```



Evaluate rpart complexity parameter cp

```
train_fit$fit$fit$fit$cptable
```

```
##   CP nsplit rel error xerror xstd
## 1  0      0         1      0    0
```

Predictions on train_set

```
treepred = predict(train_fit, train_set, type = "class")
head(treepred)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
## 3 No
## 4 No
## 5 No
## 6 No
```

Confusion matrix on train_set

```
confusionMatrix(treepred$.pred_class, train_set$failure, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
```

```
##           Reference
## Prediction    No   Yes
##        No  11062  2968
##        Yes     0     0
##
##                Accuracy : 0.7885
##                  95% CI : (0.7816, 0.7952)
##     No Information Rate : 0.7885
##     P-Value [Acc > NIR] : 0.5049
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.0000
##             Specificity : 1.0000
##          Pos Pred Value :    NaN
##          Neg Pred Value : 0.7885
##              Prevalence : 0.2115
##          Detection Rate : 0.0000
##    Detection Prevalence : 0.0000
##       Balanced Accuracy : 0.5000
##
##        'Positive' Class : Yes
##
```

Predictions on test_set

```
treepred_test = predict(train_fit, test_set, type = "class")
head(treepred_test)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
## 3 No
## 4 No
## 5 No
## 6 No
```

Confusion matrix on test_set

```
confusionMatrix(treepred_test$.pred_class, test_set$failure, positive = "Yes
")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No  Yes
##        No  4741 1272
```

```
##        Yes    0    0
##
##                Accuracy : 0.7885
##                  95% CI : (0.7779, 0.7987)
##     No Information Rate : 0.7885
##     P-Value [Acc > NIR] : 0.5075
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.0000
##             Specificity : 1.0000
##          Pos Pred Value :    NaN
##          Neg Pred Value : 0.7885
##              Prevalence : 0.2115
##          Detection Rate : 0.0000
##    Detection Prevalence : 0.0000
##       Balanced Accuracy : 0.5000
##
##        'Positive' Class : Yes
##
```

Create folds

```
set.seed(234)
folds = vfold_cv(train_set, v=5)
```

Build classification tree by R tuning

```
train_recipe2 = recipe(failure ~ loading, train_set) %>%
  step_dummy(all_nominal(), -all_outcomes())

tree_model2 = decision_tree(cost_complexity = tune()) %>%
  set_engine("rpart", model = TRUE) %>%
  set_mode("classification")

tree_grid2 = grid_regular(cost_complexity(),
                          levels = 25)

train_wflow2 =
  workflow() %>%
  add_model(tree_model2) %>%
  add_recipe(train_recipe2)

tree_res2 =
  train_wflow2 %>%
  tune_grid(
    resamples = folds,
    grid = tree_grid2
```

```
  )

tree_res2
```

```
## # Tuning results
## # 5-fold cross-validation
## # A tibble: 5 × 4
##   splits               id    .metrics          .notes
##   <list>               <chr> <list>            <list>
## 1 <split [11224/2806]> Fold1 <tibble [75 × 5]> <tibble [0 × 4]>
## 2 <split [11224/2806]> Fold2 <tibble [75 × 5]> <tibble [0 × 4]>
## 3 <split [11224/2806]> Fold3 <tibble [75 × 5]> <tibble [0 × 4]>
## 4 <split [11224/2806]> Fold4 <tibble [75 × 5]> <tibble [0 × 4]>
## 5 <split [11224/2806]> Fold5 <tibble [75 × 5]> <tibble [0 × 4]>
```

Maximize accuracy

```
best_tree2 = tree_res2 %>%
  select_best(metric = "accuracy")

best_tree2
```

```
## # A tibble: 1 × 2
##   cost_complexity .config
##             <dbl> <chr>
## 1         0.00133 pre0_mod20_post0
```

Finalize workflow

```
final_wf2 =
  train_wflow2 %>%
  finalize_workflow(best_tree2)
```

Fit the data for train_set

```
final_fit2 = fit(final_wf2, train_set)

tree2 = final_fit2 %>%
  pull_workflow_fit() %>%
  pluck("fit")
```

Predictions on train_set

```
treepred2 = predict(final_fit2, train_set, type = "class")
head(treepred2)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
## 3 No
```

```
## 4 No
## 5 No
## 6 No
```

Confusion matrix for R tuned train_set

```
confusionMatrix(treepred2$.pred_class, train_set$failure, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##        No  11062  2968
##        Yes     0     0
##
##                Accuracy : 0.7885
##                  95% CI : (0.7816, 0.7952)
##     No Information Rate : 0.7885
##     P-Value [Acc > NIR] : 0.5049
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.0000
##             Specificity : 1.0000
##          Pos Pred Value :    NaN
##          Neg Pred Value : 0.7885
##              Prevalence : 0.2115
##          Detection Rate : 0.0000
##    Detection Prevalence : 0.0000
##       Balanced Accuracy : 0.5000
##
##        'Positive' Class : Yes
##
```

Predictions on test_set

```
treepred_test2 = predict(final_fit2, test_set, type = "class")
head(treepred_test2)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
## 3 No
## 4 No
## 5 No
## 6 No
```

Confusion matrix on test_set after R tuning

```
confusionMatrix(treepred_test2$.pred_class, test_set$failure, positive = "Yes
")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No  Yes
##        No  4741 1272
##        Yes    0    0
##
##                Accuracy : 0.7885
##                  95% CI : (0.7779, 0.7987)
##     No Information Rate : 0.7885
##     P-Value [Acc > NIR] : 0.5075
##
##                   Kappa : 0
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.0000
##             Specificity : 1.0000
##          Pos Pred Value :    NaN
##          Neg Pred Value : 0.7885
##              Prevalence : 0.2115
##          Detection Rate : 0.0000
##    Detection Prevalence : 0.0000
##       Balanced Accuracy : 0.5000
##
##        'Positive' Class : Yes
##
```

Fine tuning not performed, as “Yes” values for failure are dominated by “No” values